

Report 1: Empirical Evaluation of Feedforward Architectures on MNIST

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1 Introduction and Structural Framework

This document outlines the numerical behavior of multi-layer feedforward neural networks optimizing handwritten digit classifications. The underlying process focuses on analyzing the impact of hidden layer dimensionalities when using non-linear transformation mappings.

2 Mathematical Formalism

Let $x \in \mathbb{R}^{784}$ represent an unrolled vector mapping an image structure. The mathematical computation routing across consecutive dense operations behaves according to the following iterative relation:

$$z = Wx + b$$

$$a = \tanh(z)$$

Where W corresponds to the linear transformation weights matrix and $\tanh(\cdot)$ functions as our bounding activation mechanism.

3 Computational Metrics

The numerical results generated via the Python training loop demonstrate that expanding the hidden matrix volume guarantees a more flexible boundaries optimization, steadily improving the overall accuracy on unseen test environments.